

8	(a)	(i)	Photoelectric effect is a phenomenon in which electrons are emitted from the surface of a metal when it is irradiated with electromagnetic radiation of high enough frequency.	
		(ii)	<u>Experimental observations which support the particulate nature of em radiation are</u> 1. There is a minimum frequency below which no photoelectric emission of electrons is possible, even with very intense radiation. 2. The maximum KE of the emitted electrons increases with the frequency of the radiation. The max KE does not depend on the intensity of the radiation. 3. Photoelectrons are emitted almost immediately when radiation was incident; no time lag was observed. 4. The rate of emission of photoelectrons is proportional to the intensity of the incident radiation. Max three marks.	
	(b)	(i)	For a point source of constant power, intensity is inversely proportional to the distance from the source, i.e. $I \propto 1/r^2$ $I' / I = (1/r'^2) / (1/r^2) = (r/r')^2$ $I' / 0.20 = (5/20)^2$ $I' = 0.013 \text{ W m}^{-2}$	
		(ii)	Energy of each photon $= hf$ $= 6.63 \times 10^{-34} \times 4.55 \times 10^{14}$ $= 3.02 \times 10^{-19} \text{ J}$ <u>Estimate</u> area of writing paper = $0.15 \times 0.30 \text{ m}^2$ Intensity = $E / t A = Nh f / t A = (N/t) h f / A$ $0.013 = (N/t) 3.02 \times 10^{-19} \text{ J} / 0.15 \times 0.30$ $N/t = 1.94 \times 10^{15}$	